ONE FOLDER PER JOB, ONE DIRECTION FOR THE DATA.

Raw datasets go in, are never modified, and flow one way through **cleaning → models → results**. Code lives in one Python package (**awc**), every path comes from one config file, and the README logs each finished step.

1 FOLDER TREE

```
README.md          overview + progress log, updated every step
requirements.txt   Python dependencies (all free)
pyproject.toml     installable package, test config

data/
├─ raw/            original datasets, read-only, not in git
│  ├─ mm-fit/      21 sessions · 3.8 GB
│  ├─ ec3d/        371 labelled reps · 3D
│  └─ own_recordings/   phone + Galaxy Watch 7
├─ interim/        synced + normalized (Parquet)
└─ processed/      model-ready tensors

src/awc/           the project's Python package
├─ config.py       every path and constant
├─ data/           loaders: MM-Fit, EC3D, ours
├─ features/       sync, skeleton, windows, reps
├─ models/         Model 1 · Model 2 · Model 3
├─ training/       training loops
├─ evaluation/     metrics, benchmarks (RQ1–RQ4)
├─ streaming/      Kafka producers + consumer
├─ bigdata/        Spark batch jobs
└─ coach/          live voice feedback

scripts/analysis/  first measurements on the data
notebooks/         exploration
tests/             data and code checks

experiments/       one folder per run: config + metrics
models/            trained checkpoints
reports/           briefing, figures, final report
presentation/      validation web app
docs/              course documents, notes
```

data · code · results · communication

2 HOW DATA FLOWS

`data/raw/`	MM-Fit · EC3D · our recordings, as downloaded
↓ `features/ + bigdata/ (Spark)`	
`data/interim/`	streams synced on the 30 fps clock, skeletons normalized
↓ `features/windows, reps`	
`data/processed/`	3 s windows (Model 1), one tensor per rep (Models 2, 3)
↓ `models/ + training/ (PyTorch)`	
`models/ · experiments/`	checkpoints, configs, metrics of every run
↓ `evaluation/`	
`reports/`	benchmarks answering RQ1–RQ4

Live mode: **streaming/** (Kafka) feeds the trained models and **coach/** speaks the result.

3 ROADMAP

0	Proposal & validation presentation	DONE
1	Structure, git, dataset loaders, tests	DONE
2	Data pipeline: sync, normalize, windows	NEXT
3	Model 1: recognition + rep counting	MM-FIT
4	Model 2: form check	EC3D
5	Own recordings + Kafka capture	OURS
6	**Model 3: fatigue & baseline**	CONTRIBUTION
7	Live coach with voice feedback	LIVE
8	Evaluation & benchmarks	RQ1–RQ4
9	Final report and demo	END

4 RULES WE FOLLOW

Raw data is read-only
Every result can be rebuilt from the original files.

One source of paths
All code imports its paths from `config.py`.

Data stays out of git
Only code, docs and results are versioned; `data/README.md` says how to download.

Tested and logged
Tests check the datasets; the README logs every finished step.